



US012418737B2

(12) **United States Patent**
Nishizawa et al.

(10) **Patent No.:** **US 12,418,737 B2**
(45) **Date of Patent:** **Sep. 16, 2025**

(54) **IMAGE SENSOR CAPABLE OF ADDING
PIXEL SIGNALS, CONTROL METHOD
THEREOF, AND ELECTRONIC APPARATUS**

(58) **Field of Classification Search**
CPC H04N 25/78; H04N 25/772; H04N 25/60;
H04N 25/616; H03M 1/124
See application file for complete search history.

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(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 167 days.

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(21) Appl. No.: **18/356,373**

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(22) Filed: **Jul. 21, 2023**

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(65) **Prior Publication Data**

US 2024/0040284 A1 Feb. 1, 2024

(30) **Foreign Application Priority Data**

Jul. 27, 2022 (JP) 2022-119835

(51) **Int. Cl.**

H04N 25/78 (2023.01)
H03M 1/12 (2006.01)
H04N 25/60 (2023.01)
H04N 25/772 (2023.01)

(52) **U.S. Cl.**

CPC **H04N 25/78** (2023.01); **H03M 1/124**
(2013.01); **H04N 25/60** (2023.01); **H04N**
25/772 (2023.01)

(57) **ABSTRACT**

An image sensor comprises: a plurality of sample-and-hold
circuits that sample and hold a plurality of signals input from
a plurality of different pixels; a plurality of converters that
convert the signals held in the plurality of sample-and-hold
circuits to currents corresponding to potentials of the sig-
nals, respectively; a switch that connects/disconnects the
plurality of converters; and a plurality of output circuits
wherein, in a case where the switch connects the plurality of
converters, the currents converted by the plurality of con-
verters are added and output from one of the plurality of
output circuits, and, in a case where the switch disconnects
the plurality of converters, the currents converted by the
plurality of converters are output from the plurality of output
circuits, respectively.

12 Claims, 9 Drawing Sheets

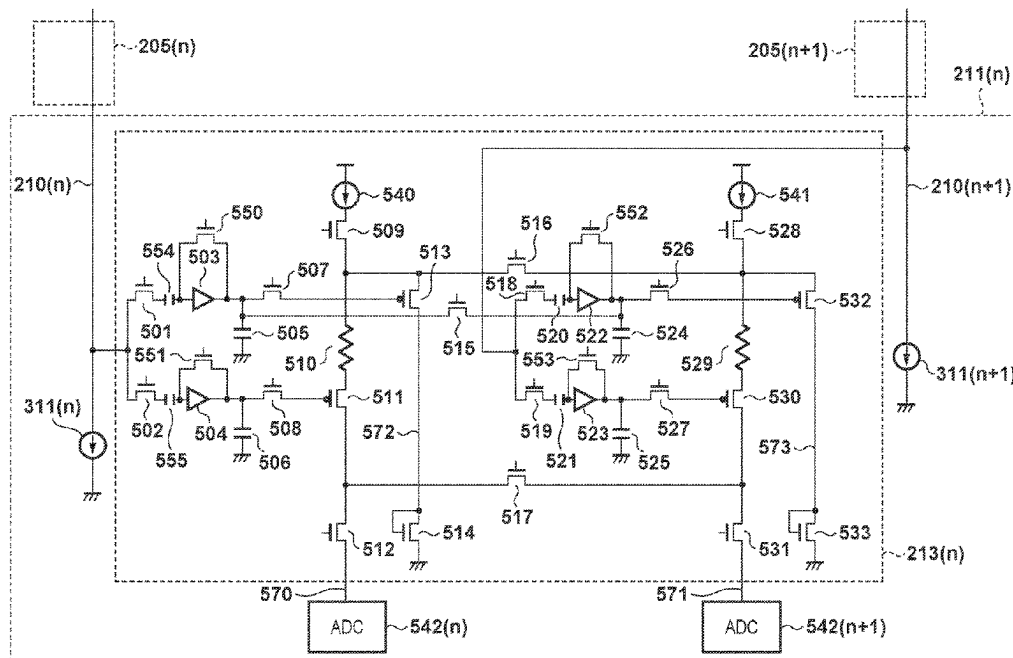


FIG. 1

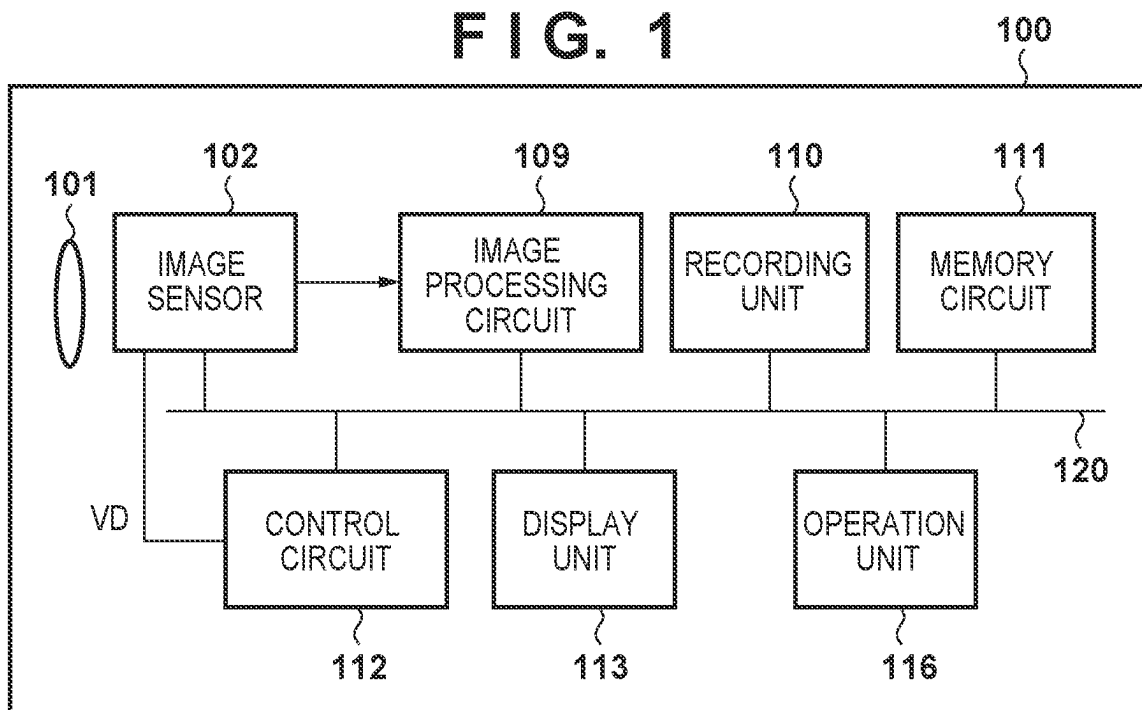


FIG. 2

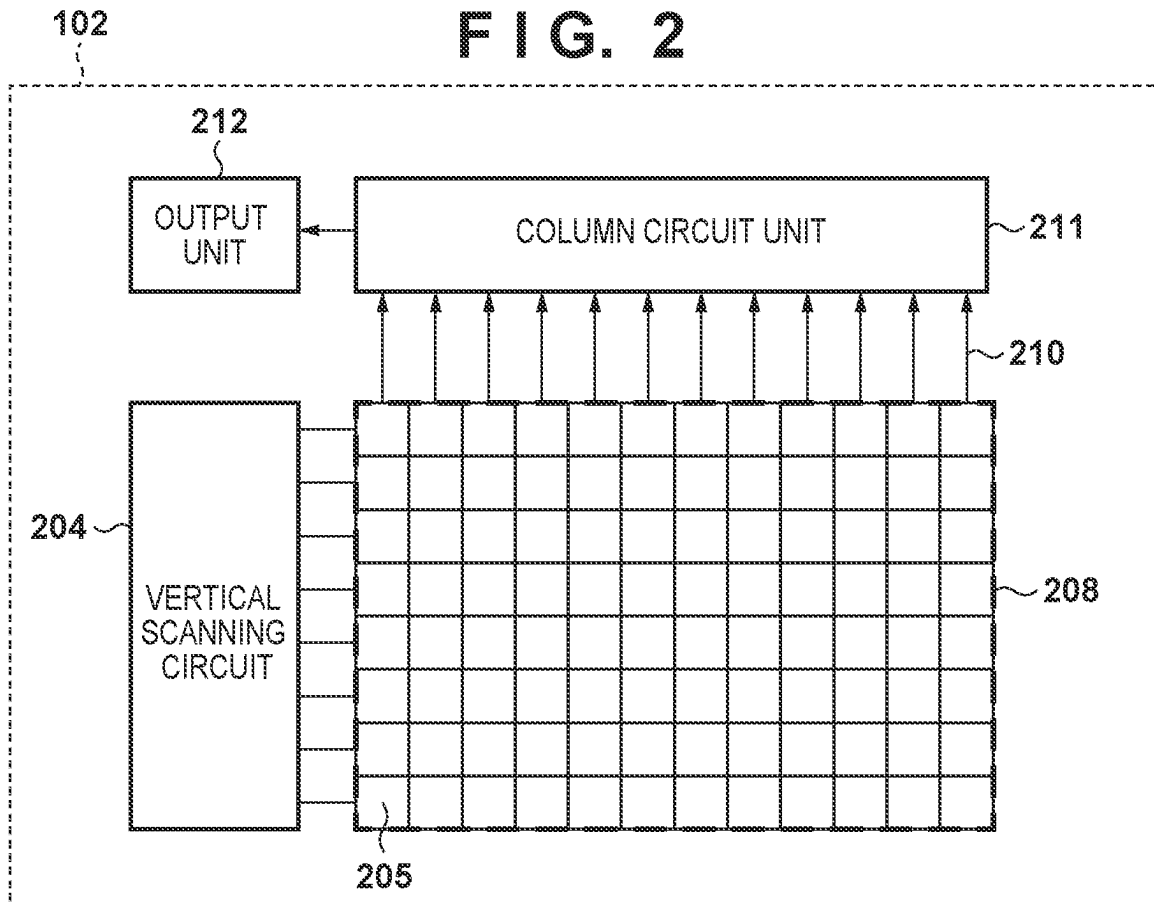


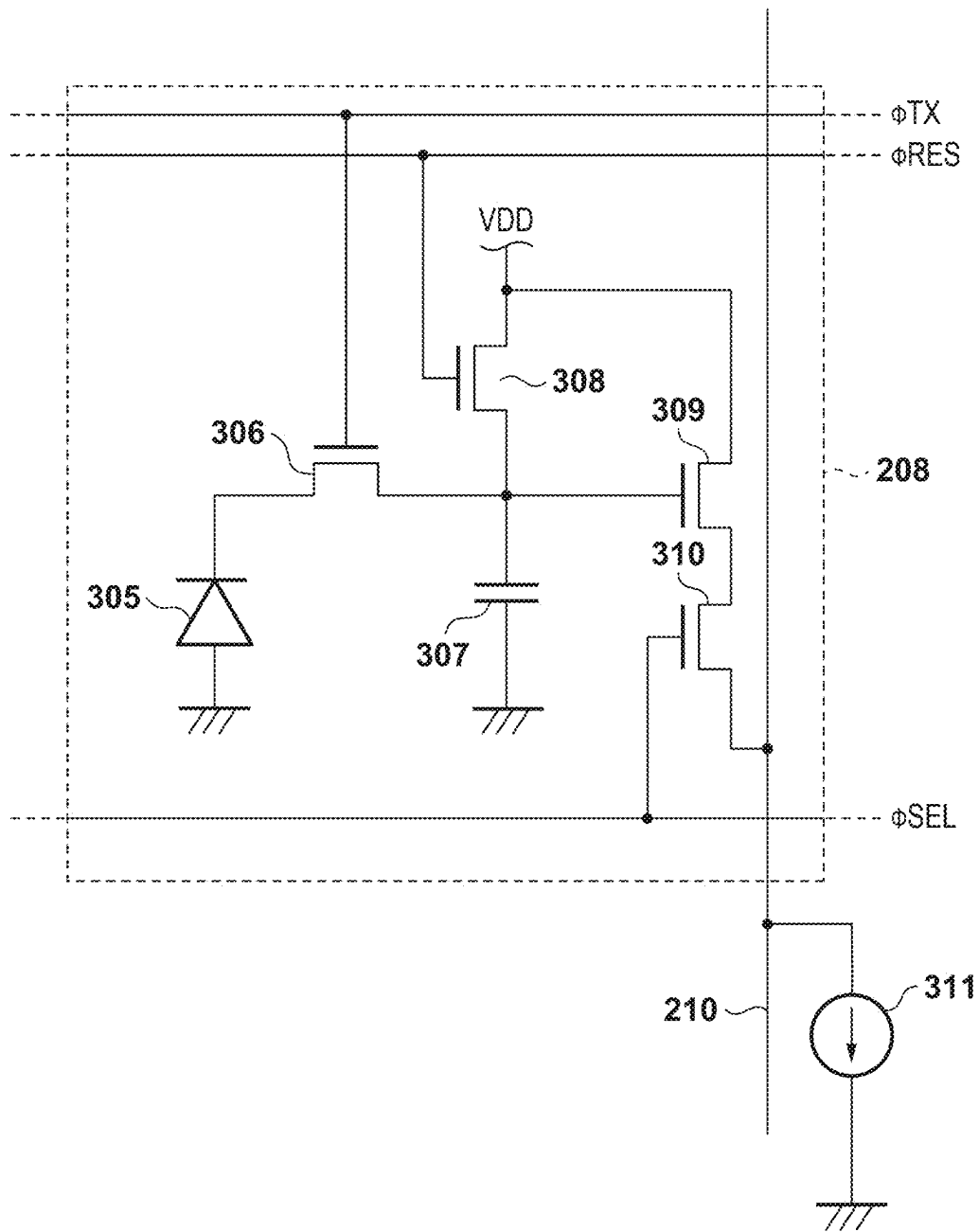
FIG. 3

FIG. 4A

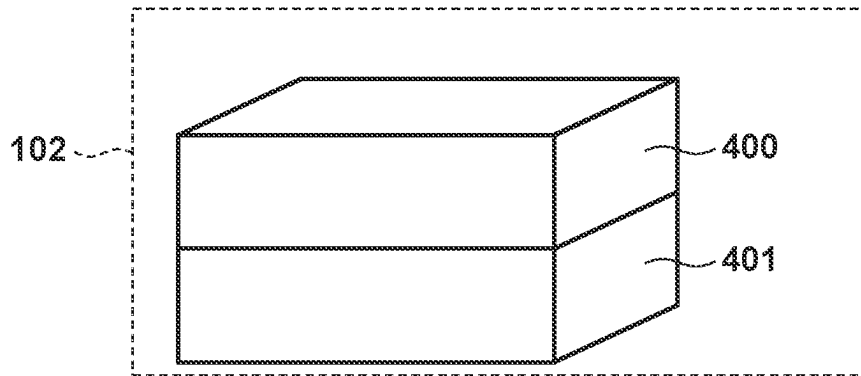


FIG. 4B

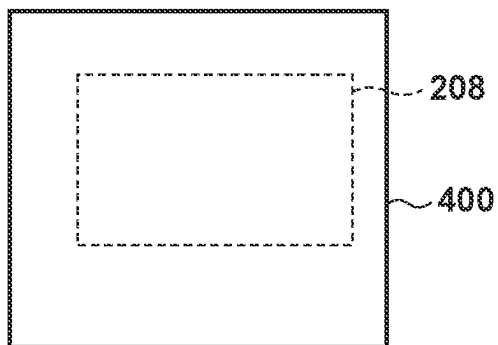


FIG. 4C

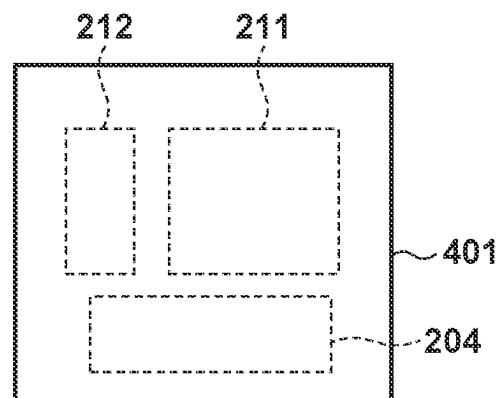


FIG. 5

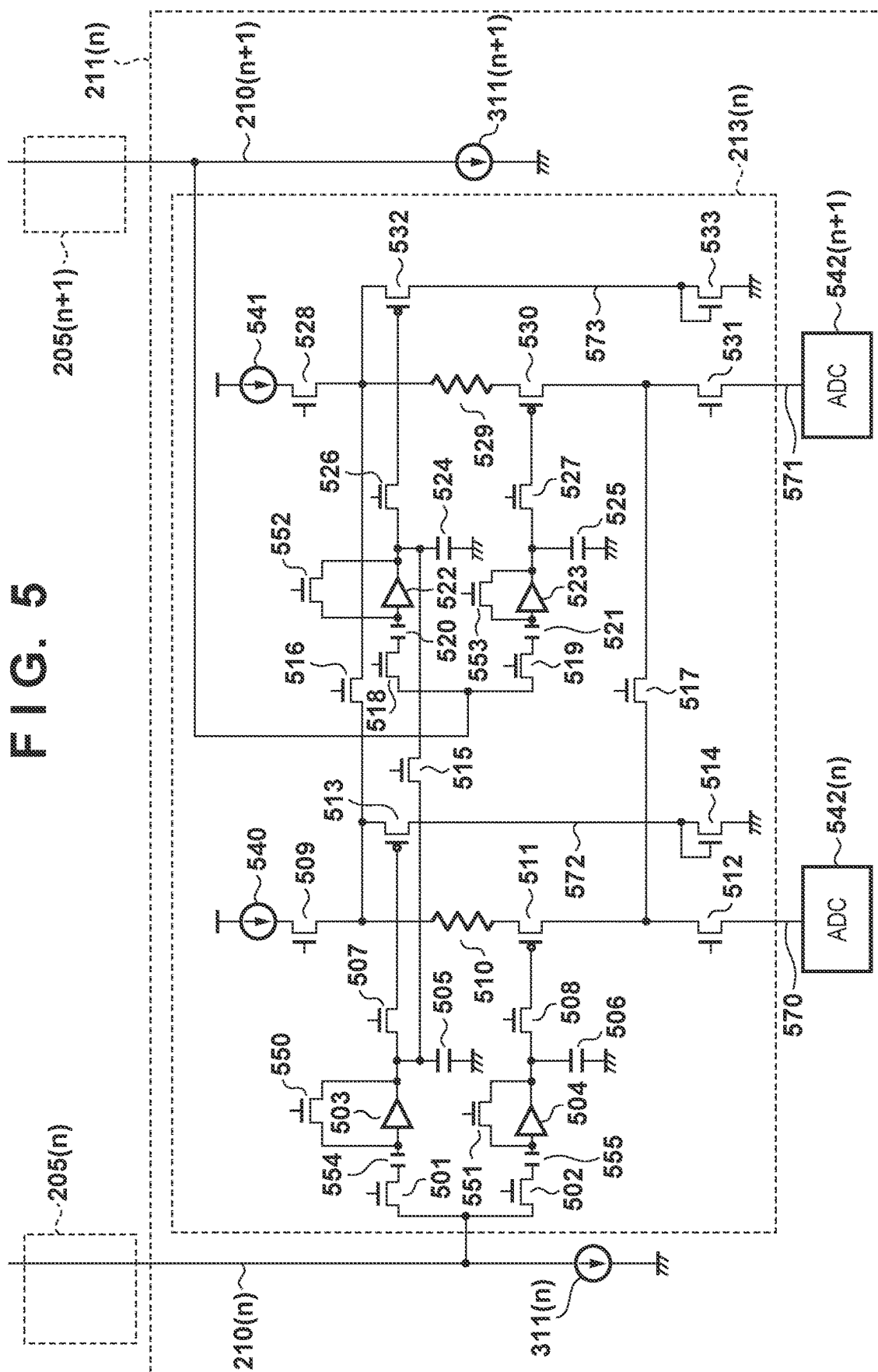


FIG. 6

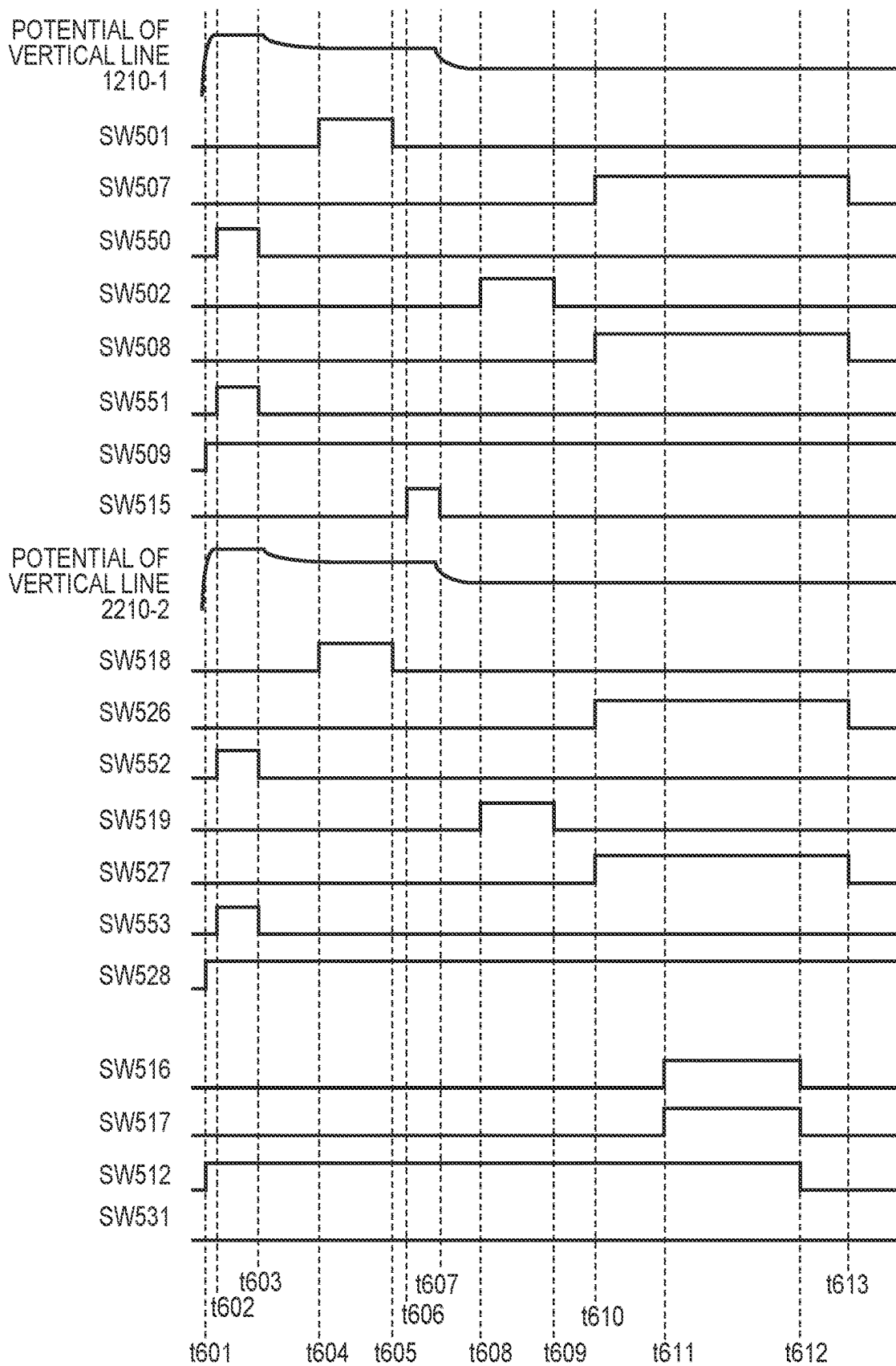


FIG. 7

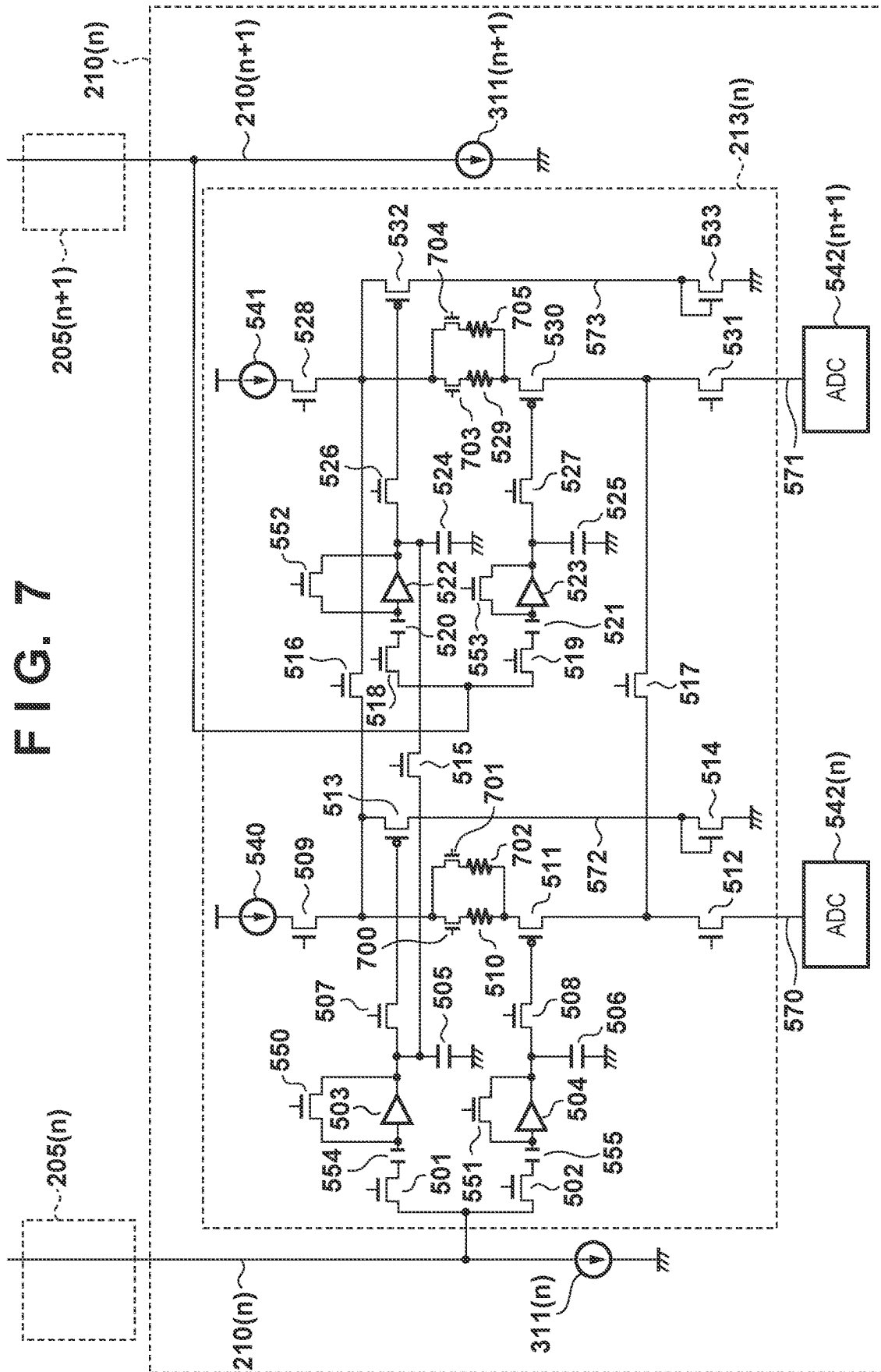


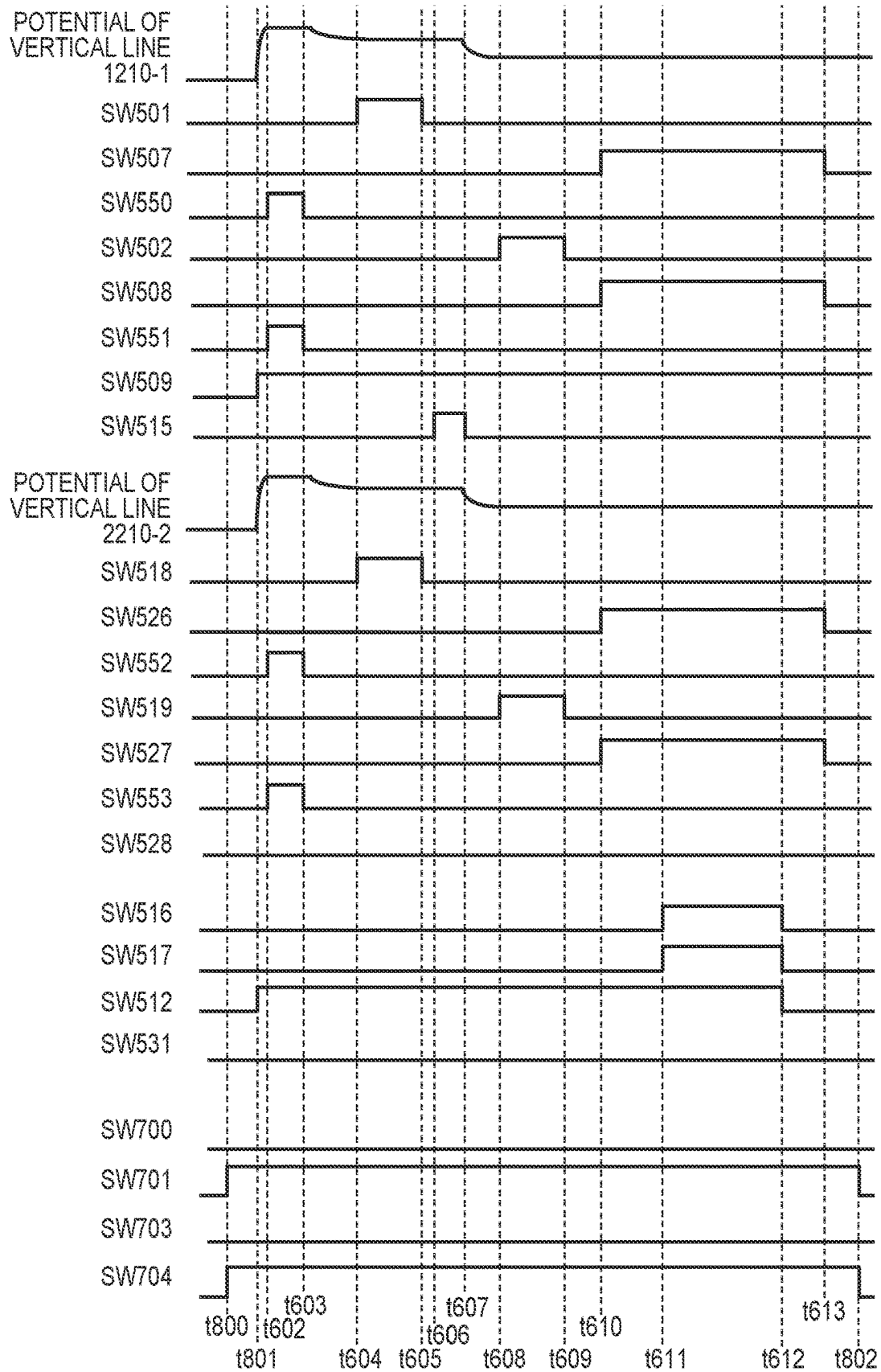
FIG. 8

FIG. 9

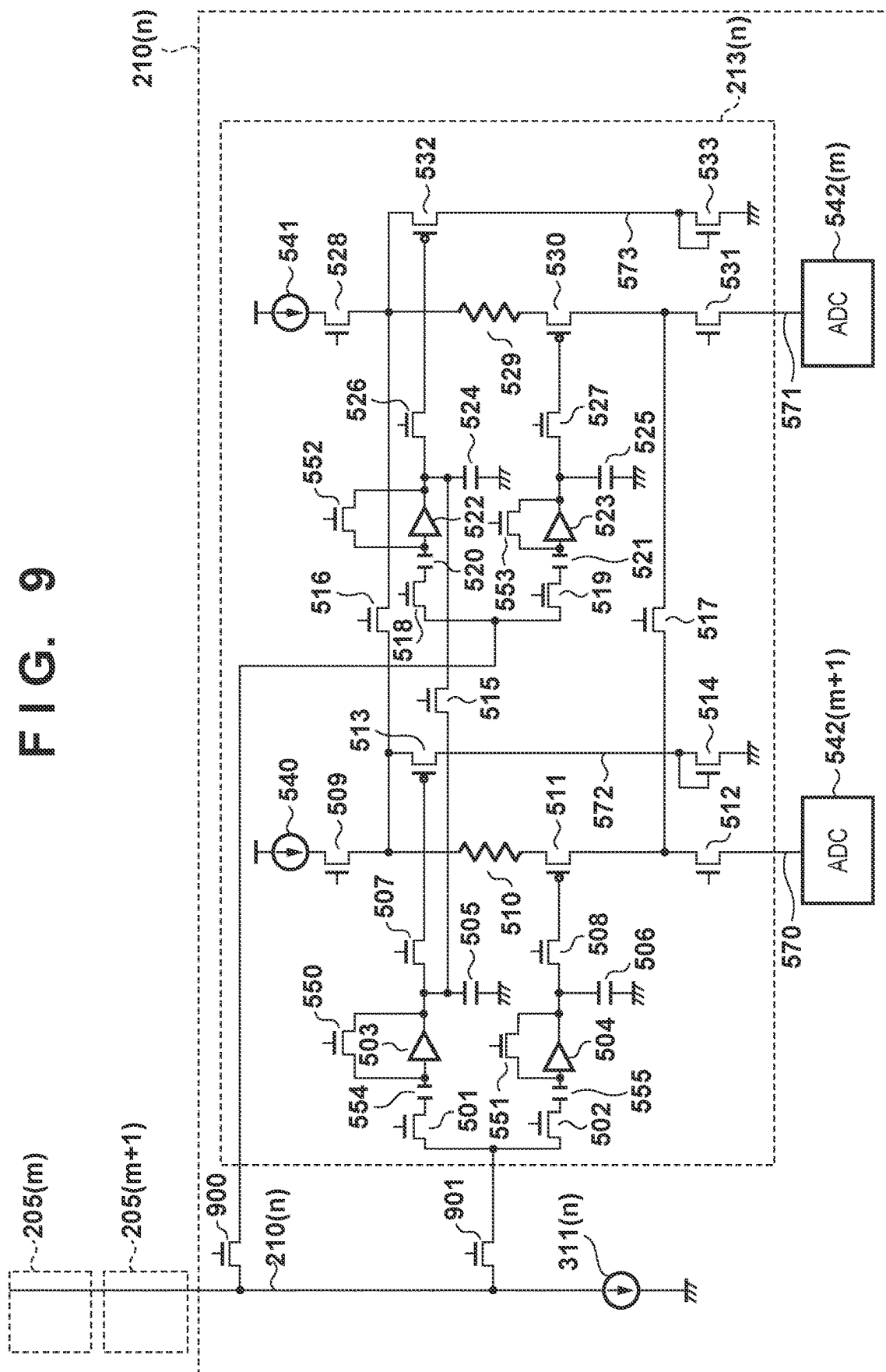
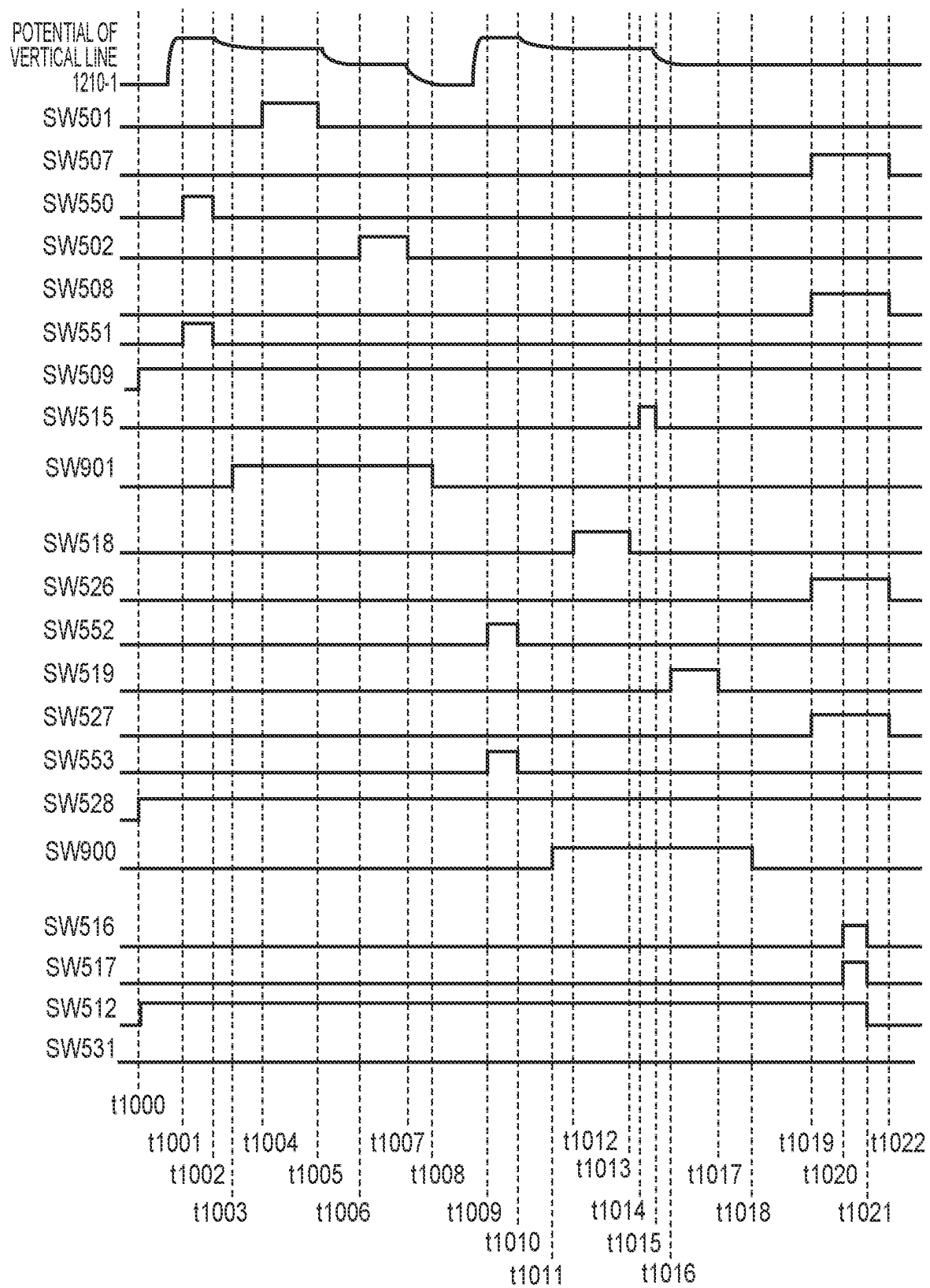


FIG. 10

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IMAGE SENSOR CAPABLE OF ADDING PIXEL SIGNALS, CONTROL METHOD THEREOF, AND ELECTRONIC APPARATUS

BACKGROUND OF THE INVENTION

Field of the Invention

The present invention relates to an image sensor, control method thereof, and electronic apparatus, and more particularly to a technique for adding pixel signals within the image sensor.

Description of the Related Art

Conventionally, as a configuration for reading a signal from each pixel of an image sensor, there is a configuration in which a signal is sequentially output to a vertical output line shared by a plurality of pixels in the same column, and which has a sample-and-hold unit that stores in a capacitor a signal corresponding to a change in the voltage of the vertical output line due to the output signal. This sample-and-hold unit samples and holds a reset signal and an optical signal, and outputs the potential of the difference between these signals as a signal from the image sensor as it is, or outputs it after performing analog/digital (A/D) conversion inside the image sensor.

Japanese Patent No. 4807440 discloses a structure having a circuit for charging and holding the voltage of a pixel output signal by a sample-and-hold circuit.

However, the Japanese Patent No. 4807440 outputs a sampled and held signal of each pixel as it is, and does not describe a method of adding signals of a plurality of pixels in the horizontal or vertical direction. Further, although it is possible to convert the signal of each pixel into a digital signal and then add the digital signals, in that case, the power consumed in the image sensor cannot be reduced.

SUMMARY OF THE INVENTION

The present invention has been made in consideration of the above situation, and adds signals of a plurality of pixels in an image sensor while suppressing the influence of noise and reducing power consumption.

According to the present invention, provided is an image sensor comprising: a plurality of sample-and-hold circuits that sample and hold a plurality of signals input from a plurality of different pixels; a plurality of converters that convert the signals held in the plurality of sample-and-hold circuits to currents corresponding to potentials of the signals, respectively; a switch that connects/disconnects the plurality of converters; and a plurality of output circuits wherein, in a case where the switch connects the plurality of converters, the currents converted by the plurality of converters are added and output from one of the plurality of output circuits, and, in a case where the switch disconnects the plurality of converters, the currents converted by the plurality of converters are output from the plurality of output circuits, respectively.

Further, according to the present invention, provided is an electronic apparatus comprising: an image sensor comprising: a plurality of sample-and-hold circuits that sample and hold a plurality of signals input from a plurality of different pixels; a plurality of converters that convert the signals held in the plurality of sample-and-hold circuits to currents corresponding to potentials of the signals, respectively; a switch that connects/disconnects the plurality of converters;

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and a plurality of output circuits wherein, in a case where the switch connects the plurality of converters, the currents converted by the plurality of converters are added and output from one of the plurality of output circuits, and, in a case where the switch disconnects the plurality of converters, the currents converted by the plurality of converters are output from the plurality of output circuits, respectively; and a processor that processes the signal output from the image sensor.

Furthermore, according to the present invention, provided is a control method of an image sensor comprising: controlling a plurality of sample-and-hold circuits to hold a plurality of signals input from a plurality of different pixels; controlling a plurality of converters to convert the signals held in the plurality of sample-and-hold circuits to currents corresponding to potentials of the signals, respectively; controlling a switch to connect/disconnect the plurality of converters; and controlling, in a case where the switch connects the plurality of converters, to add and output the currents converted by the plurality of converters, and, in a case where the switch disconnects the plurality of converters, to output the currents converted by the plurality of converters independently.

Further features of the present invention will become apparent from the following description of exemplary embodiments (with reference to the attached drawings).

BRIEF DESCRIPTION OF THE DRAWINGS

The accompanying drawings, which are incorporated in and constitute a part of the specification, illustrate embodiments of the invention, and together with the description, serve to explain the principles of the invention.

FIG. 1 is a block diagram showing a schematic configuration of an image capturing apparatus according to an embodiment of the present invention;

FIG. 2 is a schematic configuration diagram of an image sensor according to an embodiment;

FIG. 3 illustrates an equivalent circuit diagram of a pixel according to the embodiment;

FIGS. 4A to 4C illustrate schematic diagrams showing a layered structure of the image sensor according to the embodiment;

FIG. 5 illustrates a circuit diagram showing a configuration of part of a column circuit unit according to a first embodiment;

FIG. 6 illustrates a timing chart showing control at the time of addition reading according to the first embodiment;

FIG. 7 illustrates a circuit diagram showing a configuration of part of a column circuit unit according to a second embodiment;

FIG. 8 illustrates a timing chart showing control at the time of addition reading according to the second embodiment;

FIG. 9 illustrates a circuit diagram showing a configuration of part of a column circuit unit according to a third embodiment; and

FIG. 10 illustrates a timing chart showing control at the time of addition reading according to the third embodiment.

DESCRIPTION OF THE EMBODIMENTS

Hereinafter, embodiments will be described in detail with reference to the attached drawings. Note, the following embodiments are not intended to limit the scope of the claimed invention, and limitation is not made to an invention that requires a combination of all features described in the

embodiments. Two or more of the multiple features described in the embodiments may be combined as appropriate. Furthermore, the same reference numerals are given to the same or similar configurations, and redundant description thereof is omitted.

First Embodiment

[Configuration of Apparatus]

FIG. 1 is a block diagram showing a schematic configuration of an image capturing apparatus 100 according to an embodiment of the present invention.

A lens unit 101 is composed of, for example, a plurality of lenses, diaphragms, etc., and condenses light from a subject and causes it to enter an image sensor 102. The image sensor 102 photoelectrically converts incident light and outputs image data.

An image processing circuit 109 performs development processing such as color matrix processing and gamma processing on image data output from the image sensor 102. In these processes, the image processing circuit 109 causes a memory circuit 111 to hold image data as needed. The image processing circuit 109 then outputs the processed image data to a display unit 113 and/or a recording unit 110.

The display unit 113 displays image data processed by the image processing circuit 109 and other information such as shooting parameters. The display unit 113 may configure an operation unit 116 together with a touch panel, in which case an image corresponding to the operation state is displayed. The recording unit 110 records image data processed by the image processing circuit 109, information attached to the image data, and the like in a storage medium (not shown).

The operation unit 116 is composed of one or a combination of switches, dials, a touch panel, a pointing device based on line-of-sight detection, a voice recognition device, etc., and is used by the user to input various operation instructions to the image capturing apparatus 100. The operation unit 116 then generates an operation signal according to the user's operation and outputs it to a control circuit 112.

The control circuit 112 controls image processing by the image processing circuit 109 and part of actuation of the image sensor 102. It also performs control corresponding to the operation signal input from the operation unit 116 and controls data transfer to/from the memory circuit 111, display unit 113, and recording unit 110. At least part of the functions of the control circuit 112 may be performed by the image sensor 102 or the image processing circuit 109.

A bus 120 is a common path through which the control circuit 112, the image sensor 102, the image processing circuit 109, the display unit 113, the operation unit 116, the recording unit 110, and the memory circuit 111 exchange data with each other.

Although the image capturing apparatus 100 is illustrated as including the lens unit 101 in FIG. 1, the lens unit 101 may be configured so as to be detachable from the image capturing apparatus 100.

FIG. 2 is a schematic configuration diagram of the image sensor 102.

The image sensor 102 has a pixel section 208 in which a plurality of pixels 205 are arranged in two dimensions. In this embodiment, in the pixel section 208, the pixels 205 are arranged in a plurality of rows in the vertical direction and in a plurality of columns in the horizontal direction. That is, the pixels 205 are arranged in a matrix. A vertical scanning circuit 204 has a configuration for outputting various control

signals necessary for reading out signals to the pixels 205, and selects a pixel row from which signals are to be read out.

Signals read from the pixels 205 in the row selected by the vertical scanning circuit 204 are output to the vertical output lines 210 of respective columns and input to a column circuit unit 211. The column circuit unit 211 performs processing such as amplification and A/D conversion of pixel signals to generate image data. An output unit 212 sequentially outputs the image data generated by the column circuit unit 211 to the outside of the image sensor 102.

FIG. 3 shows an equivalent circuit diagram of the pixel 205.

A photodiode (PD) 305 accumulates charges generated by photoelectric conversion corresponding to the amount of incident light. The charges accumulated in the PD 305 are transferred to a floating diffusion (FD) 307 by the vertical scanning circuit 204 controlling a transfer control signal ϕ_{TX} to turn on a transfer switch 306.

An output transistor 309 constitutes a source follower circuit together with a constant current source 311 connected to the vertical output line 210, amplifies the voltage signal corresponding to the charge accumulated in the FD 307, and outputs it as a pixel signal. Here, by controlling the a selection control signal ϕ_{SEL} by the vertical scanning circuit 204, the output transistor 309 of the row whose row selection switch 310 is turned ON is connected to the vertical output line 210, and the pixel signal is output to the vertical output line 210.

Further, the vertical scanning circuit 204 controls a reset control signal ϕ_{RES} to turn on a reset switch 308, thereby resetting the FD 307 to a predetermined voltage VDD. Further, the vertical scanning circuit 204 controls the reset control signal ϕ_{RES} and the transfer control signal ϕ_{TX} to simultaneously turn on the reset switch 308 and the transfer switch 306, thereby resetting the PD 305 to the predetermined voltage VDD.

The transfer control signal ϕ_{TX} , the reset control signal ϕ_{RES} , and the row selection control signal ϕ_{SEL} are output by the control circuit 112 controlling the vertical scanning circuit 204.

FIGS. 4A to 4C are schematic diagrams showing a layered structure of the image sensor 102, FIG. 4A being an oblique view, and FIGS. 4B and 4C being top views of respective semiconductor substrates.

The image sensor 102 is formed by stacking a first semiconductor substrate 400 and a second semiconductor substrate 401 at the chip level. A region including the pixel section 208 is formed on the first semiconductor substrate 400, and high-speed logic circuits such as the vertical scanning circuit 204, the column circuit unit 211, and the output unit 212 are formed on the second semiconductor substrate 401.

FIG. 5 is an equivalent circuit diagram showing the configuration of the column circuit unit 211 for two columns in the first embodiment, and the column circuit unit 211(n) is provided for every two columns of the pixel section 208. Here, the configuration corresponding to the nth column of the pixel section 208 has (n) after the reference number indicating each configuration, and the configuration corresponding to the (n+1)th column of the pixel section 208 has (n+1) after the reference number (n is a natural number).

The column circuit unit 211(n) has a circuit 213(n) that performs sample-and-hold and outputs current, and ADCs 542(n) and 542(n+1) that perform A/D conversion. The vertical output line 210(n) and the vertical output line 210(n+1) are connected to the circuit 213(n) of the column circuit unit 211(n).

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In the above configuration, a pixel signal of the pixel **205(n)** in the row selected by the vertical scanning circuit **204** is output to the vertical output line **210(n)**, and a pixel signal of the pixel **205(n+1)** is output to the vertical output line **210(n+1)**.

[Detailed Configuration and Operation of Column Circuit Unit **211(n)**]

Independent Readout Mode

First, an independent readout mode will be explained. In the independent readout mode, pixel signals of the pixel **205(n)** and the pixel **205(n+1)** are read out separately.

First, in the column circuit unit **211(n)**, the configuration and operation of the circuit portion that samples and holds the reset signal of the pixel **205(n)** (the pixel signal output from the pixel **205(n)** after resetting) will be described.

The input terminal of a switch **501** is connected to the vertical output line **210(n)** and controlled to be ON when reading out the reset signal from the pixel **205(n)**. The output terminal of the switch **501** is connected to the input terminal of a capacitor **554**. The output terminal of the capacitor **554** is connected to the input terminal of a switch **550** and the non-inverting input terminal of a non-inverting amplifier **503**.

The output terminal of the switch **550** and the output terminal of the non-inverting amplifier **503** are connected to one terminal of a capacitor **505** and the input terminal of a switch **507**. The other terminal of the capacitor **505** is grounded. The output terminal of the switch **507** is connected to the gate of a P-type source follower **513** and the source of the source follower **513** is connected to the input terminal of a resistor **510**, and when the switch **507** is turned on, the potential of the gate of the source follower **513** becomes a potential corresponding to the reset signal.

Next, in the column circuit unit **211(n)**, the configuration and operation of the circuit portion that samples and holds the optical signal of the pixel **205(n)** (a pixel signal corresponding to the photocharge accumulated in the PD **305** of the pixel **205(n)**) will be described.

The input terminal of a switch **502** is connected to the vertical output line **210(n)** and controlled to be ON when reading out the optical signal from the pixel **205(n)**. The output terminal of the switch **502** is connected to the input terminal of a capacitor **555**. The output terminal of the capacitor **555** is connected to the input terminal of a switch **551** and the non-inverting input terminal of a non-inverting amplifier **504**.

The output terminal of the switch **551** and the output terminal of the non-inverting amplifier **504** are connected to one terminal of a capacitor **506** and the input terminal of a switch **508**. The other terminal of the capacitor **506** is grounded. The output terminal of the switch **508** is connected to the gate of a P-type source follower **511** and the source of the source follower **511** is connected to the output terminal of the resistor **510**, and when the switch **508** is turned on, the potential of the gate of the source follower **511** becomes a potential corresponding to the optical signal.

That is, the input terminal of the resistor **510** (current source **540** side) has a potential corresponding to the reset signal of the pixels **205(n)**, and the output terminal of the resistor **510** (the side opposite to the current source **540**) has a potential corresponding to the optical signal of the pixel **205(n)**.

The input terminal of the current source **540** is connected to the high potential power supply, and the output terminal is connected to the input terminal of a switch **509**. The output terminal of the switch **509** is connected to the input terminal of the resistor **510**, one terminal of a switch **516**,

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and the source of the source follower **513**. The output terminal of the resistor **510** is connected to the source of the source follower **511**. The drain of source follower **511** is connected to the input terminal of a switch **512** and one terminal of a switch **517**. The output terminal of the switch **512** is connected to an ADC **542(n)**.

In this configuration, by turning on the switches **509** and **512** and turning off the switches **516** and **517**, a current corresponding to the potential difference between the optical signal and the reset signal of the pixel **205(n)** and to the impedance of the resistor **510** flows through a current signal transfer line **570** via the switch **512** and is input to the ADC **542(n)**.

The ADC **542(n)** A/D-converts the input current signal into a digital signal. Note that the ADC **542(n)** may be of any type, such as a successive approximation type, a ramp type, or a $\Delta\Sigma$ modulation type. Also, the amount of current obtained by subtracting the current flowing through the resistor **510** from the current of the current source **540** flows to the ground through the source follower **513** and a MOS diode **514** in a residual current line **572**.

Next, in the column circuit unit **211(n)**, the configuration and operation of the circuit portion that samples and holds the reset signal of the pixel **205(n+1)** (the pixel signal output from the pixel **205(n+1)** after resetting) will be described.

The input terminal of a switch **518** is connected to the vertical output line **210(n+1)** and controlled to be ON when reading out the reset signal from the pixel **205(n+1)**. The output terminal of the switch **518** is connected to the input terminal of a capacitor **520**. The output terminal of the capacitor **520** is connected to the input terminal of a switch **552** and the non-inverting input terminal of a non-inverting amplifier **522**.

The output terminal of the switch **552** and the output terminal of the non-inverting amplifier **522** are connected to one terminal of a capacitor **524** and the input terminal of a switch **526**. The other terminal of the capacitor **524** is grounded. The output terminal of the switch **526** is connected to the gate of a P-type source follower **532**, the source of the source follower **532** is connected to the input terminal of a resistor **529**, and when the switch **526** is turned on, the potential of the gate of the source follower **532** becomes a potential corresponding to the reset signal.

Next, in the column circuit unit **211(n)**, the configuration and operation of the circuit portion that samples and holds the optical signal of the pixel **205(n+1)** (a pixel signal corresponding to the photocharge accumulated in the PD **305** of the pixel **205(n+1)**) will be described.

The input terminal of a switch **519** is connected to the vertical output line **210(n+1)** and controlled to be ON when reading the optical signal from the pixel **205(n+1)**. The output terminal of the switch **519** is connected to the input terminal of a capacitor **521**. The output terminal of the capacitor **521** is connected to the input terminal of a switch **553** and the non-inverting input terminal of a non-inverting amplifier **523**.

The output terminal of the switch **553** and the output terminal of the non-inverting amplifier **523** are connected to one terminal of a capacitor **525** and the input terminal of a switch **527**. The other terminal of the capacitor **525** is grounded. The output terminal of the switch **527** is connected to the gate of a P-type source follower **530** and the source of the source follower **530** is connected to the output terminal of the resistor **529**, and when the switch **527** is turned on, the potential of the gate of the source follower **530** becomes a potential corresponding to the optical signal.

That is, the input terminal of the resistor **529** (current source **541** side) has a potential corresponding to the reset signal of the pixel **205(n+1)**, and the output terminal of the resistor **529** (the side opposite to the current source **541**) has a potential corresponding to the optical signal of the pixel **205(n+1)**.

The input terminal of the current source **541** is connected to the high potential power supply, and the output terminal is connected to the input terminal of a switch **528**. The output terminal of the switch **528** is connected to the input terminal of the resistor **529**, one terminal of a switch **516**, and the source of the source follower **532**. The output terminal of the resistor **529** is connected to the source of the source follower **530**. The drain of source follower **530** is connected to the input terminal of a switch **531** and one terminal of a switch **517**. The output terminal of the switch **531** is connected to an ADC **542(n+1)**.

In this configuration, by turning on the switches **528** and **531** and turning off the switches **516** and **517**, a current corresponding to the potential difference between the optical signal and the reset signal of the pixel **205(n+1)** and to the impedance of the resistor **529** flows through a current signal transfer line **571** via the switch **531** and is input to the ADC **542(n+1)**.

The ADC **542(n+1)** A/D-converts the input current signal into a digital signal. Also, the amount of current obtained by subtracting the current flowing through the resistor **529** from the current of the current source **541** flows to the ground through the source follower **532** and a MOS diode **533** in a residual current line **573**.

By turning off the switches **516** and **517** (when not connected) as in the above operation, the signals of the pixel **205(n)** and the pixel **205(n+1)** can be read out independently.

Addition Readout Mode

Next, an addition readout mode will be explained. In the addition readout mode, pixel signals of the pixel **205(n)** and the pixel **205(n+1)** are added and read out.

The column circuit unit **211(n)** has the switch **516** and the switch **517** as parts for connecting circuits for sampling and holding reset signals and optical signals of the pixel **205(n)** and the pixel **205(n+1)**, respectively.

A switch **515** connects the node between the capacitor **505** and the non-inverting amplifier **503** and the node between capacitor **524** and the non-inverting amplifier **522** to average the sampling signals of the reset signals. Thereby, reset noise can be reduced by making the levels of the reset signals the same potential.

The switch **516** is a switch that connects the source of the source follower **513** and the source of the source follower **532**, and the switch **517** is a switch that connects the source of the source follower **511** and the source of the source follower **530**. By turning ON the switches **516** and **517**, current addition of the pixel **205(n)** and the pixel **205(n+1)** becomes possible.

As in the independent readout mode described above, the switches **516** and **517** are turned on while the reset signals and optical signals of the pixels **205(n)** and **205(n+1)** are sampled and held. As a result, the input terminal of the resistor **510** (current source **540** side) has a potential corresponding to the added signal of the reset signal of the pixel **205(n)** and the reset signal of the pixel **205(n+1)**, and the output terminal of the resistor **510** (the side opposite to the current source **540**) has a potential corresponding to the added signal of the optical signal of the pixel **205(n)** and the optical signal of the pixel **205(n+1)**.

In this state, by turning on the switches **509**, **512** and **528** and turning off the switch **531**, a current corresponding to the potential difference between the added signal of the optical signals of the pixels **205(n)** and **205(n+1)** and the added signal of the reset signals of the pixels **205(n)** and **205(n+1)** and to the impedance of resistor **510** flows through the current signal transfer line **570** via the switch **512** and is input to the ADC **542(n)**. The ADC **542(n)** A/D-converts the input current signal into a digital signal. Also, the amount of current obtained by subtracting the current flowing through the resistor **510** from the current of the current source **540** flows to the ground through the source follower **513** and the MOS diode **514** in the residual current line **572**.

By turning on the switches **516** and **517** (when connected) as in the above operation, the signals of the pixel **205(n)** and the pixel **205(n+1)** can be added and read out.

Timing Control During Addition

FIG. 6 is a timing chart showing the control timing of the circuit **213(n)** shown in FIG. 5 in the first embodiment, and showing the operation timing of the switches when adding and reading out the signals of pixel **205(n)** and pixel **205(n+1)**. Note that ON/OFF control of each switch shown in FIG. 6 is performed by the control circuit **112**.

First, at time **t601**, the switches **509**, **528**, and **512** are turned on to prepare to allow current to flow through the current signal transfer line **570**.

At time **t602**, the switches **550**, **551**, **552**, and **553** are turned on to reset the non-inverting amplifiers **503**, **504**, **522**, and **523**, and at time **t603**, the switches **550**, **551**, **552**, and **553** are turned off.

Between time **t603** and time **t604**, the vertical scanning circuit **204** resets the pixels **205** in the selected row. Then, when the row selection control signal ϕ_{SEL} is set to HIGH and the row selection switch **310** is turned ON, the reset signals of the FDs **307** of the pixels **205(n)** and **205(n+1)** are output to the vertical output lines **210(n)** and **210(n+1)**, respectively.

At time **t604**, the switches **501** and **518** are turned on to hold the reset signals of the pixel **205(n)** and pixel **205(n+1)** in capacitors **554** and **520**, respectively. At time **t605**, the switches **501** and **518** are turned off.

At time **t606**, the switch **515** is turned on to average the reset signals, and at time **t607**, the switch **515** is turned off.

Between time **t606** and time **t608**, the vertical scanning circuit **204** turns the transfer control signal ϕ_{TX} of the selected row to HIGH to turn on the transfer switches **306**, thereby transferring the charges accumulated in the PDs **305** of the pixels **205(n)** and pixels **205(n+1)** to the FDs **307**. At this time, since the row selection switches **310** are already turned on, the voltage signals corresponding to the charges transferred to the FDs **307** of the pixels **205(n)** and **205(n+1)** of the selected row are output to the vertical output lines **210(n)** and **210(n+1)** as optical signals, respectively.

At time **t608**, the switches **502** and **519** are turned on to hold the optical signals of the pixels **205(n)** and **205(n+1)** in the capacitors **555** and **521**, respectively. At time **t609**, the switches **502** and **519** are turned off.

At time **t610**, the switches **507**, **508**, **526**, and **527** are turned on, and the terminals of the resistors **510** and **529** are set to potentials corresponding to the reset signals and pixel signals of the pixels **205(n)** and pixels **205(n+1)**, respectively, through the source followers **513**, **511**, **532** and **530**.

At time **t611**, the switches **516** and **517** are further turned on to output a signal obtained by adding the current signals of the pixels **205(n)** and **205(n+1)** to the ADC **542(n)**. The ADC **542(n)** A/D-converts the input current signal into a digital signal.

At time **t612**, the switches **516**, **517**, and **512** are turned off, and at time **t613**, the switch **507**, **508**, **526**, and **527** are turned off, and the control ends.

As described above, according to the first embodiment, the signals output from the pixels of a plurality of columns can be converted into current signals and added with the above circuit configuration. As virtues of this, it is possible to suppress the influence of noise and to add signals of a plurality of pixels.

In the above example, the configuration in which signals of the pixels **205** are added by two columns at a time has been described, but the present invention is not limited to this. By duplicating the configuration for the (n+1)th column and connecting them with switches similar to the switches **516** and **517**, the signals output from any number of columns of pixels **205** can be converted into current signals and added.

Second Embodiment

Next, a second embodiment of the present invention will be described with reference to FIGS. 7 and 8. The second embodiment shows an example of suppressing the total amount of current by decreasing the current value of the current source and increasing the resistance value. Note that the configurations of the image capturing apparatus **100** and the image sensor **102** are the same as those described with reference to FIGS. 1 to 4C in the first embodiment, except for the configuration of the column circuit unit **211**, therefore, the description thereof is omitted.

FIG. 7 is an equivalent circuit diagram showing the configuration of the column circuit unit **211** for two columns in the second embodiment. The configuration shown in FIG. 7 is a configuration in which resistors **702** and **705** and switches **700**, **701**, **703** and **704** are added to the configuration shown in FIG. 5.

More specifically, the resistor **702** is connected in parallel with the resistor **510**, the switch **700** is connected between the resistor **510** and the switch **509**, and the switch **701** is connected between the resistor **702** and the switch **509**.

Further, the resistor **705** is connected in parallel with the resistor **529**, the switch **703** is connected between the resistor **529** and the switch **528**, and the switch **704** is connected between the resistor **705** and the switch **528**. Since the configuration other than these is the same as that described with reference to FIG. 5, description thereof will be omitted.

Here, the resistors **702** and **705** have resistance values greater than those of the resistors **510** and **529**, and the number of current sources used is reduced from two to one. With this configuration, it is possible to obtain the same signal level as in a case of using a small resistance value while suppressing the current. For example, if the resistors **510** and **529** are 100Ω , the resistors **702** and **705** with 200Ω can be used.

In the example shown in FIG. 7, a plurality of resistors are connected in parallel and one of them is selected. However, the present invention is not limited to this, and a configuration capable of changing the resistance value may be used. For example, by connecting a plurality of resistors in series and connecting to any node including both ends between the connected resistors, the resistance value can be changed. Alternatively, a variable resistor may be used.

Timing Control During Addition

FIG. 8 is a timing chart showing the control timing of the circuit **213(n)** shown in FIG. 7 in the second embodiment, and showing the operation timing of the switches when

adding and reading out the signals of a pixel **205(n)** and a pixel **205(n+1)**. Note that ON/OFF control of each switch shown in FIG. 8 is performed by the control circuit **112**. The differences from the control shown in FIG. 6 described in the first embodiment are that the control at time **t801** is different from the control at time **t601**, the control at time **t800** before time **t801** is added, and the control at time **t802** is added after the control at time **t613**.

First, at time **t800**, the switches **701** and **704** are turned on to use the resistors **702** and **705**.

Then at time **t801**, the switches **509** and **512** are turned on to prepare to allow current to flow through the current signal transfer line **570**. The switch **528** is turned on in the first embodiment, it remains off in the second embodiment.

The control from time **t602** to time **t802** is the same as the control shown in FIG. 6 of the first embodiment.

At time **t802**, the switches **701** and **704** are turned off.

It is also possible to change the addition ratio of the two signals by making the weight of the signals different. For example, by setting the resistor **702** to 200Ω and the resistor **705** to 400Ω , it is possible to add the signals from the pixel **205(n)** and the pixel **205(n+1)** at a ratio of 2:1 with the same operation.

As described above, according to the second embodiment, power consumption can be suppressed as compared with the circuit configuration shown in FIG. 5 and the control shown in FIG. 6. Also, it is possible to change the addition ratio by changing the resistance values. As a result, it is possible to correct the position of the center of gravity of the added signal in a case of reducing the number of pixels to be read out, and thus, it is possible to output a high-quality image.

Third Embodiment

Next, a third embodiment of the present invention will be described with reference to FIGS. 9 and 10. The third embodiment shows an example of current addition of signals output from a plurality of pixels **205** arranged in the vertical direction. Note that the configurations of the image capturing apparatus **100** and the image sensor **102** are the same as those described with reference to FIGS. 1 to 4C in the first embodiment, except for the configuration of the column circuit unit **211**, therefore, the description thereof is omitted.

FIG. 9 is an equivalent circuit diagram showing the configuration of the column circuit unit **211** for one column in the third embodiment. The column circuit unit **211(n)** shown in FIG. 5 has a configuration in which the signals of the pixels in the nth and (n+1)th columns are input in parallel, whereas the column circuit unit **211(n)** (n is a natural number) shown in FIG. 9 has a configuration in which the signals of the pixels **205** in the mth row and (m+1)th row of the same column (n) are input sequentially (m is a natural number). Further, the configuration in FIG. 9 is different from FIG. 5 in that switches **900** and **901** are added between the vertical output line **210(n)** and the circuit **213**. Since the configuration other than these is the same as that described with reference to FIG. 5, description thereof will be omitted.

Timing Control During Addition

FIG. 10 is a timing chart showing the control timing of the circuit **213(n)** shown in FIG. 9 in the third embodiment, and showing the operation timing of the switches when adding and reading out the signals of pixel **205(m)** in the mth row and pixel **205(m+1)** in the (m+1)th row in the nth column. Note that ON/OFF control of each switch shown in FIG. 10 is performed by the control circuit **112**.

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First, at time **t1000**, the switches **509**, **528** and **512** are turned on to prepare to allow current to flow through the current signal transfer line **570**.

Next at time **t1001**, the switches **550** and **551** are turned on to reset the non-inverting amplifiers **503** and **504**, and at time **t1002**, the switches **550** and **551** are turned off.

At time **t1003**, the switch **901** is turned on. By this time **t1003**, the vertical scanning circuit **204** performs control so that a reset signal of the FD **307** is output to the vertical output line **210(n)** by resetting the pixel **205(m)** in the selected row and turning on the row selection switch **310** of the pixel **205(m)**. Also, while the switch **901** is ON (between time **t1003** and time **t1008**), by turning on the row selection switch **310** that connects the pixel **205(m)** and the vertical output line **210(n)**, the signal of the pixel **205(m)** is transferred.

At time **t1004**, the switch **501** is turned on to hold the reset signal of the pixel **205(m)** in the capacitor **554**, and at time **t1005**, the switch **501** is turned off.

Between time **t1005** and time **t1006**, the vertical scanning circuit **204** turns the transfer control signal ϕ_{TX} of the selected row (the m th row) to HIGH to turn on the transfer switch **306**, thereby transferring the charges accumulated in the PD **305** of the pixel **205(m)** to the FD **307**. Since the row selection switch **310** of the pixel **205(m)** is ON, an optical signal is output from the FD **307** of the pixel **205(m)** to the vertical output line **210(n)**.

At time **t1006**, the switch **502** is turned on to hold the optical signal of the pixel **205(m)** in the capacitor **555**, and at time **t1007**, the switch **502** is turned off. Further, the switch **901** is turned off at time **t1008**. This completes holding of the reset signal and optical signal of the pixel **205(m)**.

Next, at time **t1009**, the switches **518** and **519** are turned on to reset the non-inverting amplifiers **522** and **523**, and at time **t1010**, the switches **518** and **519** are turned off.

At time **t1011**, the switch **900** is turned on. By this time **t1011**, the vertical scanning circuit **204** performs control so that a reset signal of the FD **307** is output to the vertical output line **210(n)** by resetting the pixel **205(m+1)** in the selected row and turning on the row selection switch **310** of the pixel **205(m+1)**. Also, while the switch **900** is ON (between time **t1011** and time **t1018**), by turning on the row selection switch **310** that connects the pixel **205(m+1)** and the vertical output line **210(n)**, the signal of the pixel **205(m+1)** is transferred.

At time **t1012**, the switch **518** is turned on and the reset signal of pixel **205(m+1)** is held in the capacitor **520**, and at time **t1013**, the switch **518** is turned off.

At time **t1014**, the switch **515** is turned on to average the reset signals of the pixel **205(m)** and the pixel **205(m+1)**, and at time **t1015**, the switch **515** is turned off.

Between time **t1015** and time **t1016**, the vertical scanning circuit **204** turns the transfer control signal ϕ_{TX} of the selected row (the $(m+1)$ th row) to HIGH to turn on the transfer switch **306**, thereby transferring the charge accumulated in the PD **305** of the pixel **205(m+1)** to the FD **307**. Since the row selection switch **310** of the pixel **205(m)** is ON, an optical signal is output from the FD **307** of the pixel **205(m+1)** to the vertical output line **210(n)**.

At time **t1016**, the switch **519** is turned on to hold the optical signal of the pixel **205(m+1)** in the capacitor **521**, and at time **t1017**, the switch **519** is turned off. Further, the switch **900** is turned off at time **t1018**. This completes holding of the reset signal and optical signal of the pixel **205(m+1)**.

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Next, at time **t1019**, the switches **507**, **508**, **526**, and **527** are turned on, and the terminals of the resistors **510** and **529** are set to potentials corresponding to the reset signals and pixel signals of the pixels **205(m)** and pixels **205(m+1)**, respectively, through the source followers **513**, **511**, **532** and **530**.

At time **t1020**, the switches **516** and **517** are further turned on to output a signal obtained by adding the current signals of the pixels **205(m)** and **205(m+1)** to the ADC **542(m)**. The ADC **542(m)** A/D-converts the input current signal into a digital signal.

At time **t1021**, the switches **516**, **517**, and **512** are turned off, and at time **t1022**, the switch **507**, **508**, **526**, and **527** are turned off, and the control ends.

In the control shown in FIG. **10**, the case of adding and reading out the signals of the pixels by two rows in the same column connected to the vertical output line **210** has been described, but the same configuration may be used to read out the signals independently. In that case, the signals of pixel **205(m)** and the pixel **205(m+1)** should be output to the ADC **542(m)** and the ADC **542(m+1)**, respectively, without turning on the switches **516** and **517** (unconnected state).

In the above example, the configuration in which the pixels **205** are added by two rows at a time has been described, but the present invention is not limited to this. By duplicating the configuration for the $(m+1)$ th row and connecting them with switches similar to the switches **516** and **517**, the signals output from any number of rows of pixels **205** can be converted into current signals and added.

In addition, in the third embodiment, an example of adding signals from the pixels **205(m)** and pixels **205(m+1)** is shown, however, it is also possible to implement a multi-sampling function that reduces noise by sampling the signal of the pixel **205(m)** in the capacitors **505**, **506**, **524**, and **525**, respectively, and adding them.

As described above, according to the third embodiment, the signals output from the pixels of a plurality of rows can be converted into current signals and added with the above circuit configuration. As virtues of this, it is possible to add signals of a plurality of pixels while suppressing the influence of noise.

In the above-described embodiments, an example in which the image sensor **102** of the present invention is mounted in the image capturing apparatus **100** has been described, however, the present invention is not limited to this, and may be mounted in various kinds of electronic devices. For example, the image sensor according to the present invention can be used in mobile communication terminals with camera functions such as mobile phones and smart phones, portable computers with camera functions, portable game machines with camera functions, and so forth.

While the present invention has been described with reference to exemplary embodiments, it is to be understood that the invention is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

This application claims the benefit of Japanese Patent Application No. 2022-119835, filed Jul. 27, 2022 which is hereby incorporated by reference herein in its entirety.

What is claimed is:

1. An image sensor comprising:

- a plurality of sample-and-hold circuits that sample and hold a first signal and a second signal output from each of a plurality of different pixels;
- a plurality of converters each of which converts a potential difference between the first signal and the second

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signal held in the plurality of sample-and-hold circuits to a current, and includes a current source and a resistor connected in series;

a switch that connects/disconnects the plurality of converters; and

a plurality of output circuits wherein, in a case where the switch connects the plurality of converters, the currents converted by the plurality of converters are added and output from one of the plurality of output circuits, and, in a case where the switch disconnects the plurality of converters, the currents converted by the plurality of converters are output from the plurality of output circuits, respectively,

wherein, among the plurality of sample-and-hold circuits, the sample-and-hold circuit that holds the first signal is connected between the current source and the resistor, and the sample-and-hold circuit that holds the second signal is connected on a side opposite to the current source with respect to the resistor.

2. The image sensor according to claim 1, wherein each of the plurality of pixels includes a photoelectric converter that photoelectrically converts incident light and generates electric charge, and

wherein the first signal is a signal output in a case where the each pixel is reset, and the second signal is a signal corresponding to the electric charge obtained by the photoelectric conversion by the photoelectric converter.

3. The image sensor according to claim 1, wherein the current source and the resistor are capable of changing their resistance values, and

wherein the resistance value of the resistor is increased in a case where the switch connects the plurality of converters.

4. The image sensor according to claim 1, wherein the current source and the resistor are capable of changing their resistance values, and

wherein the resistance values of the resistors are made different between the plurality of converters in a case where the switch connects the plurality of converters.

5. The image sensor according to claim 3, wherein the current source of one of the plurality of converters is used in a case where the switch connects the plurality of converters.

6. The image sensor according to claim 1 further comprising a pixel section in which a plurality of pixels are arranged in rows and columns,

wherein the plurality of different pixels are pixels in different columns.

7. The image sensor according to claim 6, wherein the plurality of sample-and-hold circuits, the plurality of converters, the switch and the plurality of output circuits are provided for every predetermined number of columns.

8. The image sensor according to claim 1 further comprising a pixel section in which a plurality of pixels are arranged in rows and columns,

wherein the plurality of different pixels are pixels in different rows in a same column.

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9. The image sensor according to claim 8, wherein the plurality of sample-and-hold circuits, the plurality of converters, the switch and the plurality of output circuits are provided for each column.

10. The image sensor according to claim 1 further comprising an analog-to-digital converter that converts the currents output from the plurality of output circuits into digital signals.

11. An electronic apparatus comprising:

an image sensor comprising:

a plurality of sample-and-hold circuits that sample and hold a first signal and a second signal output from each of a plurality of different pixels;

a plurality of converters each of which converts a potential difference between the first signal and the second signal held in the plurality of sample-and-hold circuits to a current and includes a current source and a resistor connected in series;

a switch that connects/disconnects the plurality of converters; and

a plurality of output circuits wherein, in a case where the switch connects the plurality of converters, the currents converted by the plurality of converters are added and output from one of the plurality of output circuits, and, in a case where the switch disconnects the plurality of converters, the currents converted by the plurality of converters are output from the plurality of output circuits, respectively; and

a processor that processes the signal output from the image sensor,

wherein, among the plurality of sample-and-hold circuits, the sample-and-hold circuit that holds the first signal is connected between the current source and the resistor, and the sample-and-hold circuit that holds the second signal is connected on a side opposite to the current source with respect to the resistor.

12. A control method of an image sensor comprising:

controlling a plurality of sample-and-hold circuits to sample and hold a first signal and a second signal output from each of a plurality of different pixels;

controlling each of a plurality of converters, that includes a current source and a resistor connected in series, to convert a potential difference between the first signal and the second signal held in the plurality of sample-and-hold circuits to a current;

controlling a switch to connect/disconnect the plurality of converters; and

controlling, in a case where the switch connects the plurality of converters, to add and output the currents converted by the plurality of converters, and, in a case where the switch disconnects the plurality of converters, to output the currents converted by the plurality of converters independently,

wherein, among the plurality of sample-and-hold circuits, the sample-and-hold circuit that holds the first signal is connected between the current source and the resistor, and the sample-and-hold circuit that holds the second signal is connected on a side opposite to the current source with respect to the resistor.

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